



Experimental study on liquid flow and heat transfer in micro square pin fin heat sink

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ABSTRACT

The heat sinks, with the total heat transfer area of $20 \times 20 \text{ mm}^2$ and an array of 625 staggered micro square pin fins of $559 \times 559 \mu\text{m}^2$ or $445 \times 445 \mu\text{m}^2$ cross section by 3 mm height were fabricated on a copper test section. Using deionized water as coolant liquid, the flow and heat transfer performance of the high pin fins were studied with the Reynolds number ranging from 60 to 800. For the $445 \times 445 \mu\text{m}^2$ cross section pin fin heat sink, the heat dissipation could reach $2.83 \times 10^5 \text{ W/m}^2$ at the flow rate of 57.225 L/h and the surface temperature of 73.4 °C. The experimental data also showed that the pressure drop and the average Nusselt number increased with the fin Reynolds number. The heat resistance decreased with, with its decreasing rate inversely proportional to, the pressure drop. Since four chosen previous correlations overestimate the flow and heat transfer performance of the present sinks, we also proposed two new correlations for the average Nusselt number and pressure drop prediction.

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1. Introduction

With the rapid development of semiconductor industry, the scale of heat dissipation rate from electronic devices increases drastically. Nowadays, the average power dissipation is about 100 W/cm^2 , and the maximum local power dissipation could reach 500 W/cm^2 [1], which is unreachable by traditional direct air cooling methods due to the maximum heat dissipation capability limitation. Since Tuckerman and Pease [2] proposed the first micro channel heat sinks and exhibited the large heat dissipation ability of micro channels, numerous investigations have been conducted on the convection heat transfer in micro channels [3–7]. Due to the small width and height, most flow patterns in micro channels were laminar. The mass transfer between the layers could be ignored so the heat transfer mode in the fluid interior is considered as heat conduction only. When the channel length is short or the flow velocity is high, there would not be enough time for interior fluid heat transfer. Some investigations have employed different methods to increase the interior layers mixing and heat transfer [6,8].

Micro scale pin fins (staggered or in-line) are the common geometry used to increase the surface area and the passage flow turbulence. A number of researchers have studied the heat transfer as well as the flow friction in these areas and presented a lot of experimental data and relational formulas [9–13]. According to the results of these formulations, the average heat transfer coefficient in short fins is larger than in long tubes. Using LIGA (Lithography Electroforming Micro Molding) micromachining process,

Marques and Kelly [14] fabricated a micro pin fin heat exchanger and tested its performance experimentally. The results showed that the micro fin heat exchangers offered the potential to control the surface temperature in high heat flux applications more effectively. Kosar et al. [15] experimentally investigated the pressure drop and friction factors in different aspect ratio micro pin fins and proposed a modified friction factor correlation. Peles et al. [16] theoretically and experimentally studied the heat transfer and pressure drop phenomena over a bank of micro pin fins with a focus on the effect of geometrical and thermo-hydraulic parameters on the total thermal resistance. They found that the micro pin fin heat radiator could take away 790 W/cm^2 when the pressure drop was two atmospheric and the surface wall temperature increment was 30.7 °C. Compared with the micro channels, the heat transfer performance of micro fins was better while the pressure drop was much bigger. Siu-ho et al. [17] experimentally investigated the pressure drop and heat transfer characteristics of a single-phase copper micro pin fin heat sink. The results indicated that the micro pin fin heat sink could be very effective at meeting the needs of high-heat-flux electronic cooling applications. The previous friction factor and heat transfer correlations might not be usable for situations beyond their original ranges of validity. Therefore, new predictive tools specifically tailored for the single-phase flow and heat transfer in the micro pin fin heat sink might be required. Qu et al. [18–20] compared thermal-hydraulic performance of a single-phase micro pin fin heat sink against a micro channel heat sink. He found that the micro-channel heat sink had a higher convection thermal resistance with a lower pressure drop at high cooling flow rates. In their study, the experiments were conducted at 30 °C and 60 °C coolant inlet temperature and

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Nomenclature

A	area of heat surface (m^2)	S_T	transverse distance of pin fins (m)
A_1	bottom areas of heat sink (m^2)	T	temperature ($^{\circ}\text{C}$)
c_p	specific heat (J/kg K)	u	flow velocity (m s^{-1})
D_c	equivalent diameter of a fin pitch (m)	W	width (m)
D_e	equivalent diameter of a fin size (m)	W_c	channel width (m)
f	friction factor	W_{fin}	fin width (m)
H	height of pin fin (m)	Δp	pressure drop (Pa)
h	heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)		
k	thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)		
L	length of pin fin region (m)		
l	distance between the thermocouple and down surface of heat sink (m)		
m	mass flux (kg s^{-1})		
N	row number of pin fins		
N_t	the number of pin fins		
Nu	Nusselt number		
p	pressure drop (Pa)		
p_d	inlet/outlet pressure drop (Pa)		
P_{fin}	Perimeter of micro pin fin cross-section (m)		
Pr	Prandtl number		
Q	total heat dissipation (W)		
q	heat flux (W m^{-2})		
Q_f	volume flow rate (L/h)		
Re	Reynolds number based on the fin size		
Re_c	Reynolds number based on the channel size		
R_{th}	heat resistance ($\text{W }^{\circ}\text{C}^{-1}$)		
S_L	streamwise distance between pin fins (m)		

Greek symbols

m_{fin}	fin efficiency coefficient
η_{fin}	fin efficiency
μ	viscosity (Pa s)
ξ	fin/channel ratio
π	3.1415
ρ	density (kg m^{-3})

subscripts

w	surface
cu	copper
f	fluid
fin	square fin
i	thermocouple ($i = 1-4$)
in	inlet
m	average
max	maximum
min	minimum
out	outlet

with six mass velocities for each inlet temperature to test the characters of micro pin fin. Their data showed that the pressure drop was over predicted by previous correlations, and the average Nusselt number increased with the average Reynolds number. They developed new correlations to predict the pressure drop and the heat transfer in the micro pin fin heat sink, based on the thermal performance, hydraulic performance, and the cost of manufacturing.

Jasperson et al. [21] also compared the micro channel and micro pin fin heat sinks. It is found that the average convection thermal resistance decreased with the flow rate for the micro pin fin heat sink, while it was almost independent on the flow rate in the micro channel heat sink. The pressure drop in the micro pin fin heat sink was larger than that in the micro channel heat sink when the flow rate was fixed.

An effective design and a performance assessment of a micro pin fin heat sink require a fundamental understanding of the fluid flow and heat transfer aspects of micro pin fins arrays. Most existing studies were empirical due to the complex nature of the fluid flow and heat transfer in micro pin fins. Therefore, many corrections were proposed to provide valuable insight into their flow and heat transfer aspects of micro pin fins. However, no one correction is conclusive.

The heat transfer and pressure drop performance of pin fins could be affected by many factors. However, pin fin height-to-diameter ratio, operating condition might be the most important ones. It is clear that, for micro pin fins, the larger the flow rate, the lower convection thermal resistance. In engineering applications, in sake of higher heat dissipation, a larger flow rate and higher power pump are needed. As we known, the pressure drop would be smaller if we get higher micro pin fins. This paper expands the micro pin fin studies by investigating the pressure drop and heat transfer in staggered micro square high pin fins. The objectives are: 1, to provide new heat transfer and pressure drop data for

liquid single-phase flow in intermediate pin-fins (pin-fin height-to-diameter ratio of 6) with $559 \times 559 \mu\text{m}^2$ or $445 \times 445 \mu\text{m}^2$ cross section by 3 mm height; 2, to develop new heat transfer and pressure drop correlations for the micro pin fins and to compare with the previous.

2. Description of the experimental setup and test section

2.1. Experimental setup

The schematic diagram of experimental setup, which consists of a cooling system, a heating and control unit, a data collecting & processing unit, is shown in Fig. 1. In the cooling system, a low temperature sink was used to keep the inlet temperature of the experimental section at a fixed value. Deionized water, which was driven from the low temperature sink, flowed through a ripple damper and a 50 μm filter, and then was shunted into two parts by three way valves. The one part passed through the flowmeter, and later flew into the micro pin fin test section. After taking away the heat dissipation, it mixed with the other part and returned to the low temperature sink.

The heating and control unit was designed to heat the test section. The key part of this unit was the heat source simulator. The heat source simulator was made of a pure copper block and heated by eight 300W power cartridge heaters which were inserted from the bottom of the copper block (Fig. 2). Those cartridge heaters were controlled by an adjustable power meter. The top of copper block was trapped to $20 \times 20 \times 20 \text{ mm}$ cube, while the lower part of copper is 80 mm in diameter and 110 mm in height. To minimize heat loss and to keep the 1-D heat transfer characteristic in the direction perpendicular to the top surface, all the surfaces other than the top one were insulated by Aspen Aerogels insulation (thermal conductivity, $k = 0.012 \text{ W/m K}$). This heating and control

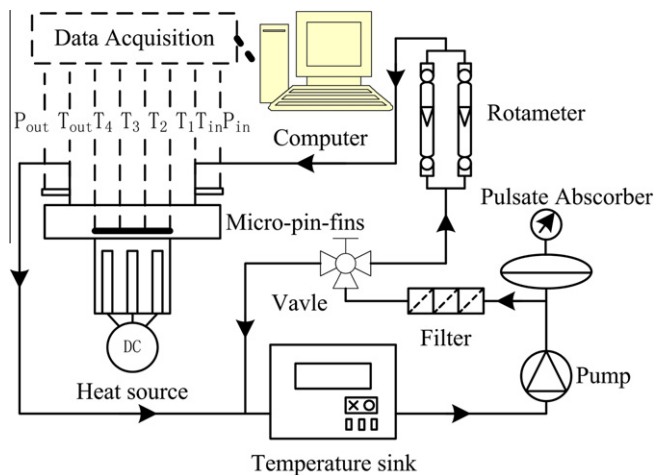


Fig. 1. Schematic diagram of experimental setup.

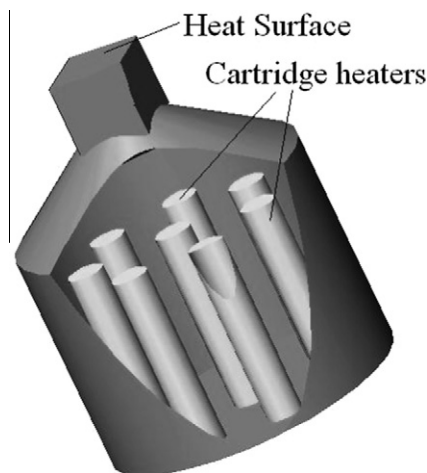


Fig. 2. Heat source simulator.

unit can supply the different constant heat flux by shift adjustable power meter.

The data collecting & processing unit, composed of pressure transducers, thermocouples, an Agilent 34970A digital acquisition/control system, flowmeters and a personal computer, was adopted to collect the temperature and pressure data in the micro pin fins heat sink.

2.2. Micro pin fin heat sink test section

The micro pin fins heat sink (shown in Fig. 3), which consisted of a cover plate, a housing, a micro pin fins heat sink and a 1-D conduction cube, had a top area of $2 \times 2 \text{ cm}^2$. Two types of square micro pin fins heat sinks with an array of 625 staggered micro square pin fins in 3 mm height were fabricated on the top of pure coppers. The top view of the micro square pin fins is shown on the left side of Fig. 3. The dimension of micro-pin-fin was measured by SEM (scanning electron microscopy). Fig. 4 shows the SEM pictures of the two heat sinks. The dimension based on the SEM measurements are given in Table 1. Below the micro square pin fin heat sink, four type-T thermocouples, which were named T_1 – T_4 , were uniformly inserted into the cube along the flow direction. On the bottom side of the micro pin fin heat sink, a 1-D conduction cube was made to reduce the temperature errors. Before being fabricated, the heater was numerically studied using a three-dimensional heat diffusion model with the software FLUENT. It was found that the heater exhibited a good one-dimensional heat transfer characteristic [22].

The cover plate was also made by pure copper to avoid the fluid leakage due to different material thermal stresses. The housing contained both inlet and outlet tanks of square micro pin fin arrays. Two type-T thermocouples were fixed in the inlet and outlet tank for temperature collection. Two pressure transducers were placed on the inlet/outlet tubes for pressure data measurement. The cover plate and the housing were bolted on the top of the micro pin fins to form a closed flow passage for the water. To make the experimental data more accurately, the surface of square micro pin fins heat sink was also insulated by Aspen Aerogels insulation (thermal conductivity, $k = 0.012 \text{ W/m K}$).

3. Data reduction and uncertainty analysis

3.1. Flow friction

According to the structures of micro pin fin heat sink, the maximum flow velocity in fin channel is obtained by

$$u_{\max} = \frac{Q_f}{A_{\min}}, \quad (1)$$

where $A_{\min}$ is the minimum transverse flow area of square micro pin fins which was calculated by:

$$A_{\min} = WH \frac{W_c / \sin 45^\circ}{2S_T}. \quad (2)$$

The Reynolds number based on the minimum channel width is given by:

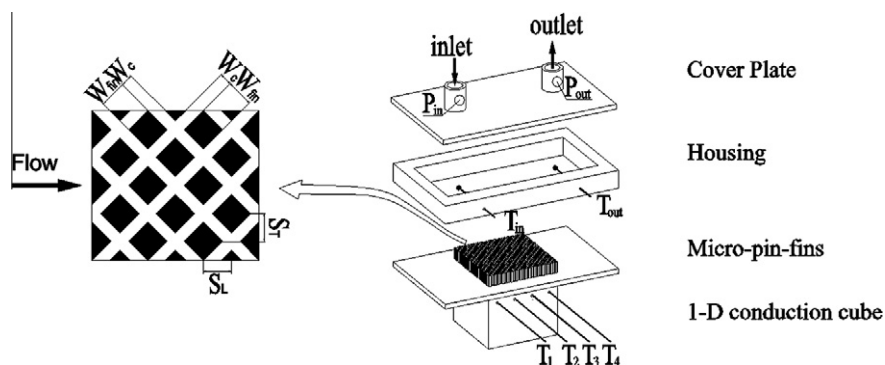


Fig. 3. The structure of square micro pin fin heat sink.

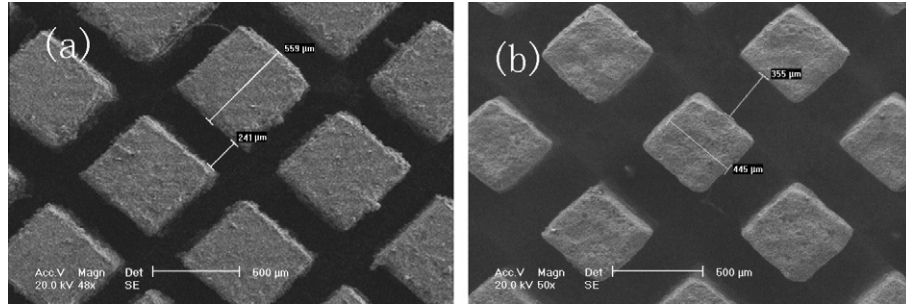


Fig. 4. SEM pictures of two square micro pin fins arrays. (a) $559 \times 559 \mu\text{m}^2$ cross section (type 1) (b) $445 \times 445 \mu\text{m}^2$ cross section (type 2).

Table 1

Dimensions of micro square pin fin heat sink (Unit: mm).

No.	Length (L)	Width (W)	Height (H)	C width (W_c)	F width (W_{fin})	S_T	S_L
Type 1	20	20	3	0.241	0.559	0.5657	0.5657
Type 2	20	20	3	0.355	0.445	0.5657	0.5657

$$\text{Re}_c = \frac{\rho u_{\max} D_c}{\mu_f} \quad (3)$$

Based on Eq. (3), the Reynolds numbers are the same for the two heat sinks when the flow rate is fixed.

The minimum channel width is determined by

$$D_c = \frac{2HW_c/\sin 45^\circ}{H + W_c/\sin 45^\circ} \quad (4)$$

The pressure transducers are set on the inlet/outlet of heat sinks, the pressure drop measured by these transducers is composed of the inlet/outlet pressure loss (p_d) and channel pressure drop (Δp). Therefore, the channel pressure drop is calculated by:

$$\Delta p = p_{in} - p_{out} - p_d \quad (5)$$

where the inlet/outlet pressure drop is obtained by:

$$p_d = 0.15 \rho u_{in}^2 + 0.1 \rho u_{out}^2 \quad (6)$$

$$u_{in} = \frac{Q_f}{0.002^2 \pi} \quad (7)$$

$$u_{out} = \frac{Q_f}{HW} \quad (8)$$

The flow friction is calculated by [23]:

$$f = \frac{\Delta p}{N(\frac{\rho u_{\max}^2}{2})} \quad (9)$$

N is the row number of pin fins.

The maximum ratio of fin/channel is:

$$\xi = \frac{2S_T}{W_c/\sin 45^\circ}.$$

3.2. Heat transfer

The heat dissipation by the micro square-pin-fins is related to the temperature difference between inlet and outlet, which were measured by thermocouples.

$$Q = c_p m (T_{out} - T_{in}) \quad (10)$$

and the heat flux is expressed as following:

$$q = \frac{Q}{A_1} = \frac{c_p m (T_{out} - T_{in})}{A_1} \quad (11)$$

The average temperature of water and the temperature of channel bottom are defined as:

$$T_m = \frac{T_{in} + T_{out}}{2} \quad (12)$$

and

$$T_w = \frac{T_1 + T_2 + T_3 + T_4}{4} - \frac{ql}{k_{cu}} \quad (13)$$

The average heat transfer coefficient of micro square pin fins is evaluated from:

$$h_{ave} = \frac{Q}{(A_1 - N_t W_{fin}^2 + \eta_{fin} N_t A_{fin})(T_w - T_m)}, \quad (14)$$

where η_{fin} represents the fin efficiency:

$$\eta_{fin} = \frac{\tanh(m_{fin} H_{fin})}{m_{fin} H_{fin}} \quad (15)$$

and m_{fin} is the fin efficiency coefficient:

$$m_{fin} = \sqrt{\frac{h_{ave} P_{fin}}{k_{cu} W_{fin}^2}} \quad (16)$$

The Reynolds number and Nusselt number based on the fin dimension are calculated by the following equations respectively:

$$\text{Re} = \frac{\rho u_{\max} D_e}{\mu_f} \quad (17)$$

$$\text{Nu}_{ave} = \frac{h_{ave} D_e}{k_f} \quad (18)$$

The fin dimension is obtained by:

$$D_e = \sqrt{2} W_{fin}. \quad (19)$$

The heat resistance of heat sink is composed of two parts: the conduction resistance through the copper and the effective convection resistance. In this study, those two parts are combined to gain the total resistance:

Table 2
Experimental uncertainties.

Parameters	Uncertainty (%)	Parameters	Uncertainty (%)
Fluid temperature	2	Flow rate	4
Equivalent diameter	2	Flow velocity	6.6
Heat quantity	6.3	Friction resistance	12.8
Re	6.9	fRe_c	9.4
Nu_{ave}	10.6		

$$R_{th} = \frac{T_w - T_m}{Q}. \quad (20)$$

3.3. Uncertainty analysis

All measurements in the experiment have their maximum errors. The data acquired by the thermocouples were compared with a precision mercury thermometer with $\pm 0.1^\circ\text{C}$ rated accuracy in the range of $0\text{--}300^\circ\text{C}$, and the maximum error is 2%. The rotameter was calibrated by weighing on an electronic scale with an accuracy of 0.1 g before the experiment, and its maximum error is 4%. The accuracy of the channel width was measured by SEM and the max error is ± 0.001 mm.

According to the traditional methods of error analysis, the uncertainty can be calculated by [24]:

$$\sigma_y = \pm \sqrt{\sum_{i=1}^N \left(\frac{\partial f}{\partial (x_i)} \right)^2 \sigma_{x_i}^2}. \quad (21)$$

For example, the uncertainty of Nu_{ave} , determined by the uncertainties of Q , A_1 , W_{fin} , A_{fin} , T_w , T_m , D_e using Eqs. (14) and (18), is 10.6%.

The details of measurement errors are listed in Table 2.

4. Results and discussions

4.1. Flow friction

Fig. 5 shows relations between the pressure drop and the Re_c number of two micro pin fin heat sinks, which are qualitatively different from that of the Poiseuille flow in tubes. When the fluid flows through heat sink, it flows across fins in the heat sinks, leading to the mixing of the fluid between channels. The vortex is also evolving behind the fins, resulting in a more complex characteristics of the pressure drop in micro fin heat sinks than in micro

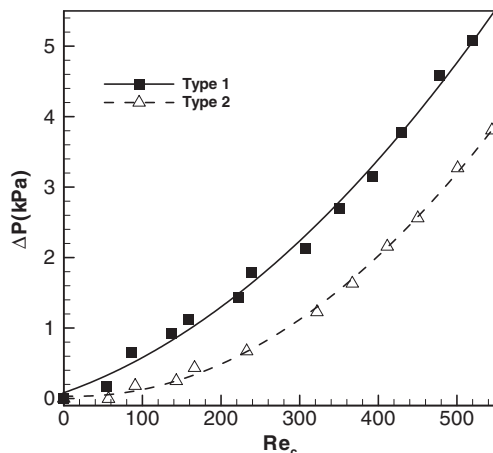


Fig. 5. The pressure drop vs Re_c number.

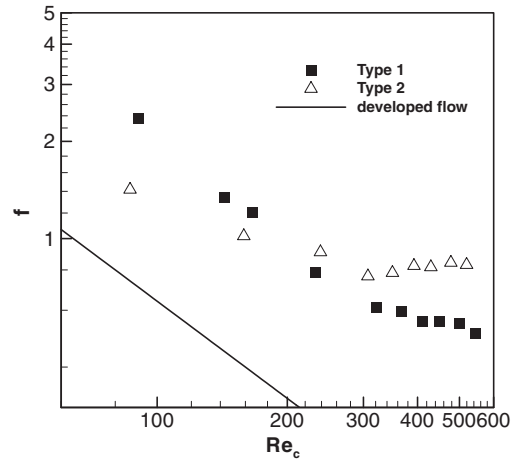


Fig. 6. The friction factor vs Re_c number.

channel heat sinks. However, the disturbed effect makes the pressure drop characteristics in micro square pin fin heat sink similar to the disturbed flow. Comparison between the type 1 and type 2 heat sinks results shows that the pressure drop is inversely proportional to the size of the fin-pitch for a fixed Reynolds number (Re_c), which is due to the fact that a smaller fin-pitch results in a faster flow velocity, and thus a stronger vortex and a higher pressure drop. For square cylinder cross flow, the vorticities across wake are affected by streamwise velocity distribution. The magnitude of the vorticity peak largely depends on the cylinder width and free stream velocity magnitude, thus, a faster flow velocity results in a stronger vortex. On the other hand, the drag due to the square cylinder is proportional to velocity square, thus, a faster flow velocity results in a higher pressure drop.

According to the definition, when the flow rate Q_f are kept the same for both heat sinks, the Re_c remains the same, while the flow friction factors are different due to the different geometry of micro pin fins. Therefore, the relationships between flow friction f and Re_c for the two heat sinks are different as shown in Fig. 6. It also shows that the laminar turbulent transition with respect to the Re_c number occurs at above 300, earlier than that in the convention tube flow, due to the staggered arrays structures in the heat sinks which cause the fluid mixing between channels and the vortex evolving behind the fins more intensively and frequently. When Re_c is less than 300, the flow friction factors of the two micro pin fin heat sinks in Fig. 6 have a similar pattern, decreasing with the Re_c , for the full development laminar flow. After Re_c crosses 300, they tend to be constants. It is very interesting to find that the friction factor of type 2 is smaller than that of type 1 for Re_c less than 200 and larger for Re_c greater than 200. For laminar flow and evenly distributed pin fins, the friction factor can be given as $\frac{180}{Re} \left(\frac{\varepsilon}{1-\varepsilon} \right)^{0.4}$, where ε is porosity [16]. As expected, the friction factor decreases with increasing of Reynolds number and porosity. For turbulent flow, Daloglu [25] found that there exists a critical spacing (the distance between two cylinder is about $1 \sim 2d$, where d is cylinder diameter) at which the pressure drop takes a minimum value, which is even less than the pressure drop obtained for the case of in the channel with single cylinder. In this study, the ratio of the neighboring pin fins distance and pin fin hydraulic diameter is about 1. Probably, this is the main reason that the friction factor of type 2 greater than that of type 1 when Re_c is larger than 200.

To predict the characteristics of flow friction in present micro square pin fin heat sinks, two correlations are proposed:

$$f = 43.322 Re_c^{-0.9116} \varepsilon^{0.9362} \quad (0 < Re_c < 300) \quad (22)$$

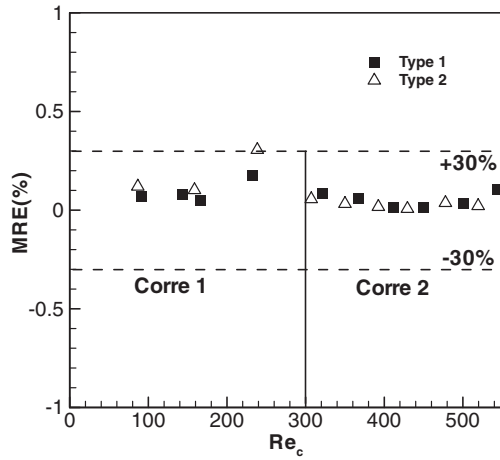


Fig. 7. The mean relative error along the Re_c number.

and

$$f = 1.6361 Re_c^{0.01076} \zeta^{-0.94496} \quad (300 < Re_c < 550), \quad (23)$$

with the first one denoted as Corre 1 and the second one Corre 2.

The mean relative error (MRE), which is used to estimate the error of the two correlations with respect to the experimental data, is defined as:

$$MRE = \frac{1}{M} \sum \frac{|f_{ave,exp} - f_{ave,pred}|}{f_{ave,exp}} \times 100\%, \quad (24)$$

where M is the total number of data points. Fig. 7 shows the comparison of the experimental data and two correlations. It is found that the maximum errors of these two correlations are 30.5% and 10.4%, respectively. In this study, the pin fin height-to-width ratio ranges from 6–8, thus, the flow in the heat sink is highly three-dimensional. On the basis of first order channel ratio, the present friction factor correlations may have pin fin height-to-width ratio limitation.

4.2. Heat transfer

The variations of q with T_w for different flow rates in the type 1 and type 2 heat sinks are shown in Fig. 8. The heat dissipation q increases with the surface temperature T_w for a fixed flow rate since the classical heat transfer theory predicts a constant heat

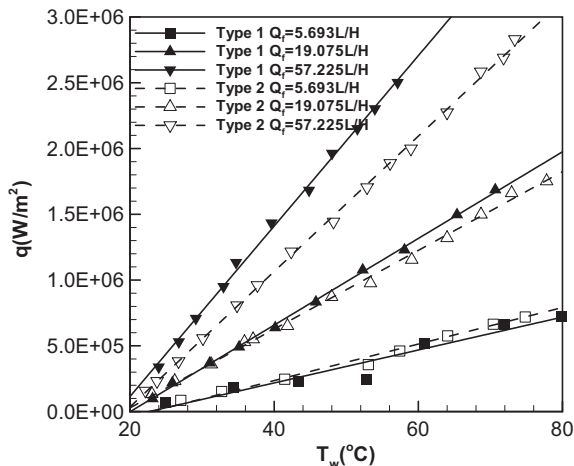


Fig. 8. The heat flux vs surface temperature.

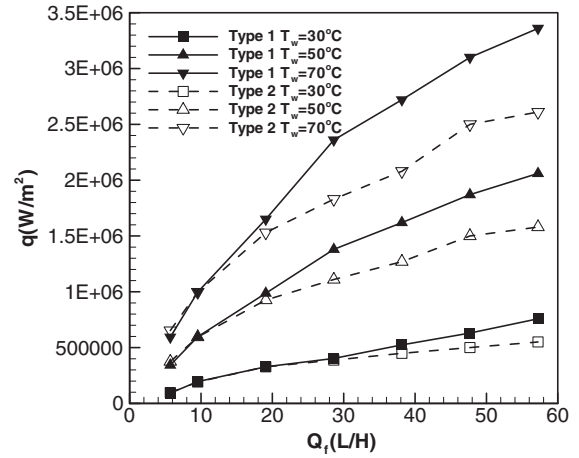


Fig. 9. The heat flux vs flow rate.

transfer coefficients for a constant flow rate in the single phase heat transfer and, thus, a stronger heat dissipation under the constant flow rate for a bigger temperature difference between the heat surface and the fluid. The heat dissipation of 2.50×10^6 W/m² was observed with the flow rate of 57.225 L/h and the surface temperature of 57.1 °C for the type 1 heat sink. Under the same flow rate condition, the heat dissipation was about 2.83×10^6 W/m² with the surface temperature of 73.4 °C for the type 2 heat sink. These observations suggest that a micro square pin fin heat sink could have a huge cooling capacity and therefore the potential to meet the demand of high power density electric devices. Due to larger heat transfer surface, it is found that heat dissipation for heat sink 1 is higher than heat sink 2.

The relation between the heat dissipation q and the flow rate Q_f for fixed surface temperatures in the type 1 and type 2 heat sinks is shown in Fig. 9. For a fixed surface temperature of a heat sink, increasing the flow rate enhances the heat dissipation, which implies the increase of the temperature gradient along fins and thus reduces the fin efficiency. Therefore, the increasing rate of the heat dissipation with respect to the flow rate decreases. Given the same surface temperature and total flow rate for both heat sinks, the type 1 heat sink is supposed to have a larger velocity than the type 2 heat sink, which implies a larger heat transfer coefficient according to the classical heat transfer theory, therefore, a larger heat dissipation. This is the reason for the difference of the heat dissipation between the type 1 and type 2 heat sinks increases with the flow

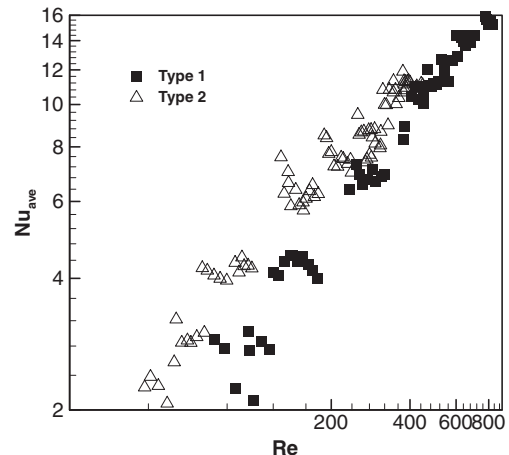


Fig. 10. The Nusselt number vs Reynolds number.

rate and the surface temperature, corroborating the finding about the heat transfer coefficient as shown in Fig. 8.

Fig. 10 plots the variation of Nu_{ave} with the Reynolds number on a log-log scale for the type 1 and type 2 heat sinks. It shows that Nu_{ave} increases with Reynolds number for both heat sinks, which might be described by power lines of $Nu_{ave} \propto Re^m$. The Nu_{ave} of the type 2 heat sink is larger than that in the type 1 when the Reynolds number is small (less than 300). However, as the Reynolds number increases, they tend to be identical. This is probably due

to the laminar to turbulent transition with the Re_c number above 300 in the two heat sinks. For the laminar flow (Re less than 300), the larger channel size of the type 1 heat sink, compared to the type 2, results in a smaller flow velocity for a fixed Reynolds number and thus a smaller heat transfer coefficient, which in turn leads to a smaller Nu_{ave} . As the Reynolds number increases to the range of turbulence, the formation of coherent structures may be shed and advected downstream. The shed structures are primarily responsible for wake mixing properties. Thus, enhanced fluid

Table 3
Correlations for heat transfer in pin fin heat sinks.

No.	Reference	Fin types	Height/width ratio	Fluid type	Reynolds number	Average Nusselt number	MRE (%)
1	Zukauskas et al. [9]	Convention	–	Air	10–100	$Nu_{ave} = 0.9Re^{0.4}Pr_{f,ave}^{0.36}(Pr_{f,ave}/Pr_{w,ave})^{0.25}$	136.6
2	Short and Raad [12]	Convention	<7.5	Air	>1000	$Nu_{ave} = 0.76Re^{0.33}Pr_{f,ave}^{1/3}(S_L/d_e)^{0.16}(S_T/d_e)^{0.2}(H/d_e)^{-0.11}$	24.1
3	Kosar et al. [15]	Micro	4.1	R-123	130–320	$Nu_{ave} = 0.0432Re^{0.99}Pr_{f,ave}^{0.21}(Pr_{f,ave}/Pr_{w,ave})^{0.25}$	121.6
4	Qu and Siu-Ho [19]	Micro	3.35	Water	45.9–179.6	$Nu_{ave} = 0.0241Re^{0.953}Pr_{f,ave}^{0.36}(Pr_{f,ave}/Pr_{w,ave})^{0.25}$	41.4
5	Present study	Micro	6–8	Water	0–800	$Nu_{ave} = 0.143Re^{0.615}Pr_{f,ave}^{1/3}$	11.0
6	Present study	Micro	6–8	Water	0–800	$Nu_{ave} = 0.1245Re^{0.6106}Pr_{f,ave}^{0.36}(Pr_{f,ave}/Pr_{w,ave})^{0.25}$	11.2

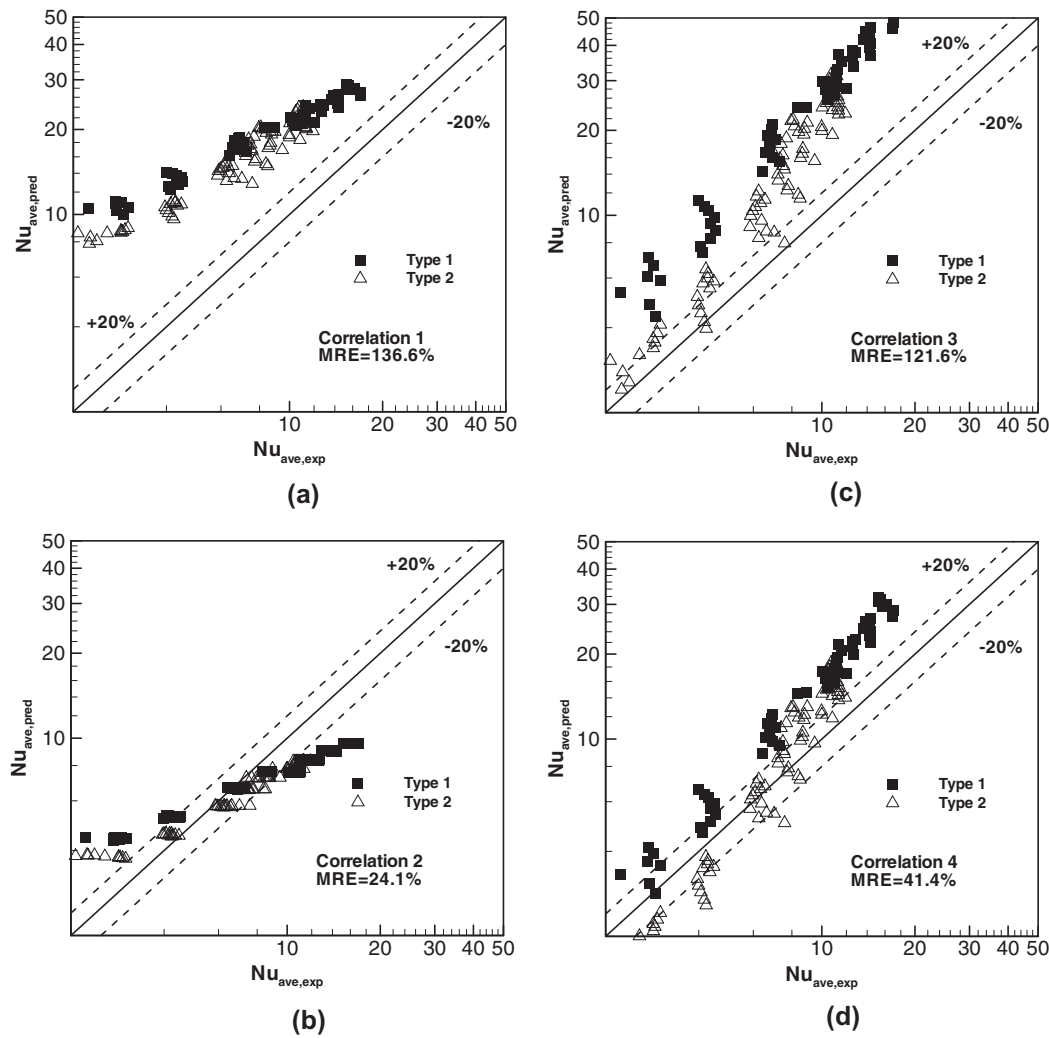


Fig. 11. Comparison of the average Nusselt number data with predictions of previous correlation.

mixing and the coherent structures effect impel the heat transfer coefficient increase. The Nu_{ave} of those two heat sinks tend to be equal.

In the classic heat transfer theory [26], the Nu_{ave} number is a function of the Reynolds number and the Prandtl number when the shape of apparatus is fixed, written as:

$$Nu_{ave} = cRe^m Pr_{f,ave}^n \quad (25)$$

and

$$Nu_{ave} = cRe^m Pr_{f,ave}^n (Pr_{f,ave}/Pr_{w,ave})^l. \quad (26)$$

$Pr_{f,ave}$ in Eqs. (25) and (26) is the water Prandtl number at T_m , and $Pr_{w,ave}$ in Eq. (26) is water Prandtl number at T_w .

There have been lots of correlations for the single-phase heat transfer in pin-fin arrays. Four previous typical correlations are selected and summarized in Table 3. the correlation 1–2 were based on the heat transfer data from conventional pin fin arrays, and the correlation 3–4 were based on the heat transfer data from micro pin fin arrays.

To compare the previous correlations with the present data, the average relative error (MRE) of Eqs. (25) and (26) are calculated as

$$MRE = \frac{1}{M} \sum \frac{|Nu_{ave,exp} - Nu_{ave,pred}|}{Nu_{ave,exp}} \times 100\% \quad (27)$$

where M is the total number of data points.

Present Nu_{ave} data are compared with previous predictions in Fig. 11(a)–(d). It is found that the MRE of the correlation 1–4 with respect to the present data ranges from 23.1%–136.6% as listed on Table 3. It should be noted that the correlations from Short et al. [12] and Zukauskas et al. [9] are for air, while from Kosar et al. [15] is for R-123. It is very interesting that Short's [12] correlation, which is for air flow at $Re > 1000$ flow condition, gives minimum MRE. The deviations increase with the Nu_{ave} number for the correlations of the convention pin fin arrays, however, decrease for those of the micro pin fin arrays. The relative variation in the slopes of the correlations in Table 3 and Fig. 11 may be due to one dimensional correlations for a highly three dimensional flow. The flow, thus heat transfer performance, is not only affected by pin fin configuration, but also by height-to-width ratio and longitudinal and transverse pitches of fin pin. As early in 1972, Zukauskas et al. [9] found that increase of m (in Eq. (25)) with a constant longitudinal and a decreasing transverse pitch. Therefore, it is concluded that the m in correlations developed for highly three dimensional flows should be a function of height-to-width ratio and longitudinal and transverse pitches of fin pin.

From Table 3 and Fig. 11, it is found that the previous heat transfer correlations may not be able to fit the present data of the micro square pin fin heat sinks. The reasons for poor prediction performance of previous heat transfer correlations include: (1) Effect of fluids property. Though the use of the Prandtl number implies a more general application, it does not guarantee that these correlations apply to general gas or liquid flows; (2) Effect of geometric configuration and pin fin spacing. Both geometric configuration and pin fin spacing affect fluid flow inside heat sinks. (3) The height-to-width ratio of diamond shaped pin fins in this study is about 6–8, the flow is highly three dimensional. That may be the main reason for different critical Reynolds and heat transfer performance. Thus, the discrepancy between the previous correlation predictions and the present data is not necessarily related to the weaknesses of the correlations themselves but maybe to the geometric parameters and operating conditions. Therefore, we propose two new heat transfer correlations to fit these types micro-square pin fin heat radiator's heat transfer performance.

In present study, the working fluid is water. Therefore, the range of prandtl number is not wide enough to change the exponents n and l of Eqs. (25) and (26). As shown in Table 3. The n

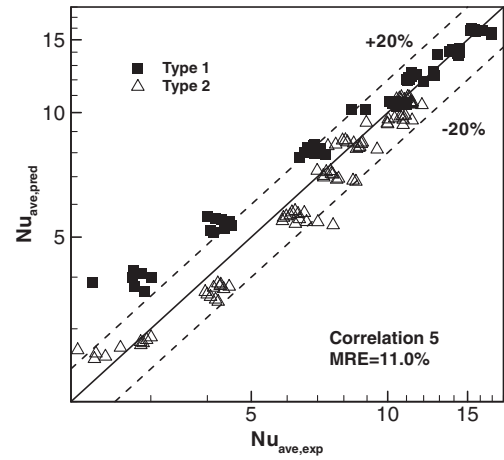


Fig. 12. Comparison of the average Nusselt number data with predictions of correlation 5.

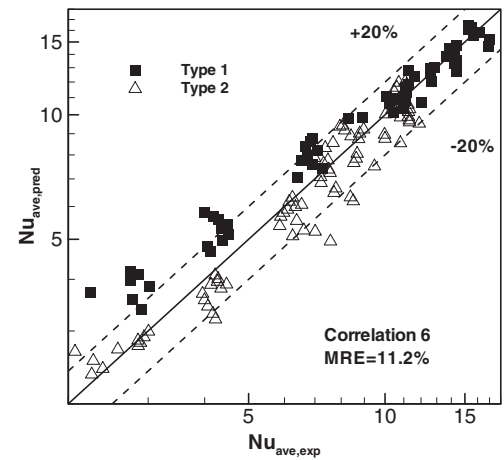


Fig. 13. Comparison of the average Nusselt number data with predictions of correlation 6.

and l in Short's [12] correlations are both $1/3$, while they are 0.36 and 0.25 in Zukauskas et al. [9] and Qu and Siu-Huo's [19] correlations. In our study, we assume n is $1/3$ in Eq. (25), while the n and l in Eq. (26) are 0.36 and 0.25, respectively. The two new correlations are

$$Nu_{ave} = 0.143Re^{0.615} Pr_{f,ave}^{1/3} \quad (28)$$

and

$$Nu_{ave} = 0.1245Re^{0.6106} Pr_{f,ave}^{0.36} (Pr_{f,ave}/Pr_{w,ave})^{0.25}. \quad (29)$$

The two correlations are also listed on Table 3 for comparison. Fig. 12 and Fig. 13 compare the new correlations with respect to the present data, which have the MRE of 11.0% and 11.2%, respectively, acceptable for experiments.

4.3. Heat resistance

To predict the performance of heat sinks, the variation of heat resistance with heat dissipation of the type 1 and type 2 heat sinks are shown in Fig. 14. It is found that when the flow rate was 47.688 L/h, the heat resistances are nearly fixed values, invariant with respect to the heat dissipation. However, distinguishable difference between the heat resistances of the two heat sinks are

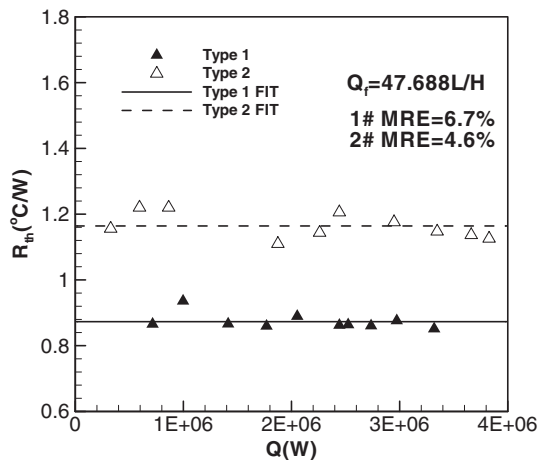


Fig. 14. The heat resistance vs the heat dissipation.

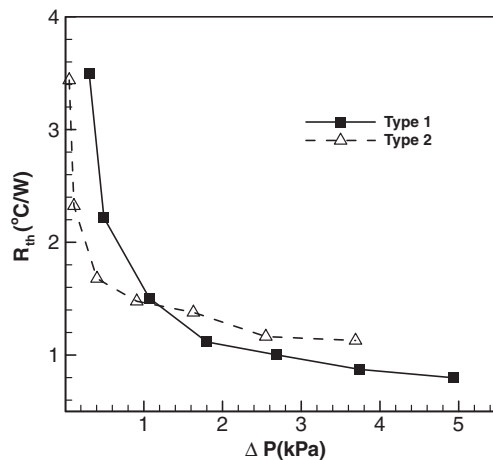


Fig. 15. The heat resistance vs the pressure drop.

observed. This is due to the same reason mentioned above, i.e., a smaller width of flow path in the type 1 resulting a larger flow velocity for a given same flow rate.

Therefore, assuming the heat resistance is invariant with respect to the heat dissipation, the heat resistance is a function of flow rate only. It is clear that the increase of the flow rate corresponds to an increase of pressure drop. To evaluate the heat transfer and pressure drop performance, the variation of heat resistance with pressure drop of different flow rate in type 1 and type 2 heat sinks is presented in Fig. 15.

For both heat sinks, the heat resistance decreases with the increase of the pressure drop. The decreasing rate is inversely proportional to the pressure drop. The type 1 heat sink has larger heat resistances for small pressure drop ($\Delta P < 1.1$ kPa), but smaller ones for large pressure drop ($\Delta P > 1.1$ kPa), than the type 2 heat sink.

As shown in Table 1, the type 1 heat sink has a flow path width (W_c) of 0.241 mm and the pin fins width (W_{fin}) of 0.559 mm, while the flow path and the pin fins width of the type 2 heat sink are 0.335 mm and 0.445 mm, respectively. For the two heat sinks, the pressure drop of 1.1 kPa corresponds to the Reynolds number of 10^2 for the type 1 and 164 for the type 2 (based on length scale W_c), respectively. This implies, with the same pressure drop, the velocity of coolant liquid in the type 2 heat sinks is faster due to the smaller flow path width (W_c). In this study, the heat resistance is an effective one, which includes the conduction resistance and

the coolant liquid convection resistance. When the pressure drop is less than 1.1 kPa, due to weak convection, the conduction resistance dominates. Hence, the heat resistance of the type 2 is smaller. When Re_c increases, the convection resistance becomes dominant. As a result, the effective resistance of type 1 becomes smaller.

It should be pointed out that the heat resistance is much bigger than that of Prasher et al. [23]. This is due to the difference in the pin-fin heights. The pin height in our study is 3 mm, larger than 0.2–0.31 mm in Prasher's study. The higher pin height, the lower the fin efficiency η_f . Hence, the enhancement in the heat transfer for micro pin fin is not necessary due to the increase of the area. As Prasher said, micro pins do more than just increasing the area.

5. Conclusions

In this study, two micro staggered square high pin fin heat sinks with different channel sizes were fabricated. Using deionized water as working fluid, the performance of pressure drop and heat transfer in staggered square long micro pin fins were experimentally studied. The main conclusions include:

- 1) For both heat sinks, the pressure drop increased with the Re_c number. The flow friction factor transition phenomenon appeared at $Re_c \sim 300$.
- 2) Both heat sinks exhibited huge heat dissipation capability. The experimental data showed that, for the type 2 heat sink, the heat dissipation could reach 2.83×10^6 W/m² at the flow rate of 57.225 L/h and the surface temperature of 73.4 °C, and, therefore, meet the demand of high power heat removal. The heat dissipation increased with the flow rate for a fixed surface temperature while the increasing rate decreased with the flow rate.
- 3) The Nusselt number increased with the fin Reynolds number. For both heat sinks, the heat transfer was over predicted by the previous correlations. Therefore, we proposed new correlations for the average Nusselt number prediction. The Nusselt number varies as $Re^{0.61}$.
- 4) The heat resistance decreased with the pressure drop. The deceleration rate was faster for the small pressure drop and slower for the large pressure drop. The type 1 heat sink has larger heat resistances for small pressure drop ($\Delta P < 1.1$ kPa), but smaller ones for large pressure drop ($\Delta P > 1.1$ kPa), than the type 2 heat sink.

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